

From pin wiggle to agents

You already built a self-checking UART testbench with clock, reset, and inline expects

Six pairs on the map

- The lab lines up six classic pieces with six UVM roles
- Pin wiggle maps to driver plus virtual interface
- Hard-coded vector lists map to sequences and a sequencer
- A display peek maps to a monitor and analysis ports, observation splits from stimulus
- Inline expect maps to a scoreboard, checks happen on transactions, not only wires
- A flat module testbench maps to an environment and agents

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Basic TB vs UVM map

Map **wiggle-the-pins** testbenches onto UVM roles: sequences, agents, drivers, monitors, and scoreboards. Starter: **UART TX byte** — pin drive already linked to the driver.

Starter example: UART TX byte with *pin wiggle* ↔ *driver* selected and marked mapped. Click other pairs, Mark mapped, or Demo all.

Load starter example

Challenges 0 / 22 cleared

Quiz: goal: This lab's main idea is...

☐ map classic pin TB roles onto UVM components

☐ compile Accellera UVM in the browser

☐ synthesize a UART VIP

☐ replace all \$display with SVA

Show hintCheckNextIdle

12345678910111213141516171819202122

Core ideas

Pins → transactions

UVM moves intent from edges to items.

Split roles

Drive, watch, and check are separate components.

Agent

Driver + monitor + sequencer around one interface.

Reuse

Factory and VIP beat copy-paste TBs.

Browser lab starter

learn_uart — tb vs uvm map

Real RTL/TB practice

- In Track A, restate the core idea in one sentence
- On paper, map these five classic pieces to UVM
- Scoreboard in a UART byte send
- You do not need a simulator for this beat, the mapping is the skill
- When you later read a real `uart_agent`

Pitfalls to watch

- Do not treat this lab as a substitute for running self-checking RTL
- Do not assume UVM removes pin knowledge
- A scoreboard is not only display, it compares predicted versus observed transactions
- Sequences are not magic, they replace the vector lists you would have pasted into tasks
- And remember: literacy first
- Factory, objections, and RAL come later

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load starter, click through all six pairs, mark each mapped
- On paper, draw test to env to agent to DUT for UART TX
- When you are ready, take the short quiz, then continue to VIP anatomy